

North South University

Department of Electrical and Computer Engineering

Project Update Report

Project Name: Classic Tic Tac Toe Game

Course Code: CSE 115

Faculty: MSRB

Section: 4

Group Project Number: Two

Group Members	ID
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Abstract:

This report presents an overview of the current design and development progress of our **Classic Tic Tac Toe Game** program developed in the C programming language. The system provides a simple, console-based interactive platform for two players to play the classic game, tracking wins and draws. This project follows a structured programming approach using arrays, loops, and conditional statements. This update describes the core features, programming techniques, and current progress of the project.

Introduction:

Tic Tac Toe is a timeless strategy game traditionally played on a 3x3 grid. Our project aims to create a **Classic Tic Tac Toe Game** featuring an interactive console interface where players can take turns marking spaces and tracking game statistics over multiple rounds.
We are making use of several fundamental programming concepts in C, such as standard input/output functions (`<stdio.h>`), character arrays for board representation, `do-while` loops for game repetition, and `for` loops for rendering and game management. Modular structuring is being implemented to keep the code clean, readable, and easy to debug.

Methodology:

The project follows a systematic programming approach combining arrays and control flow statements:
* **Data Structures:** A character array `n` of size 9 is utilized to represent the board positions (initially mapped from `1` to `9`), allowing easy updates as players make their moves. Score tracking variables (`xw`, `ow`, `dw`) are integrated to record X wins, O wins, and draws across rounds.
* **Game Loop & Control Flow:** A outer `do-while` loop allows continuous gameplay across multiple matches, while inner loops and conditional logic manage turn-taking and board display.
* **Console Interface:** Formatted `printf` statements are designed to render a clear, visual 3x3 ASCII grid directly in the console window.

Task Division:

- Main Code and Overall Project Management: S.M. Nahiyan Rahman Akib
- GitHub Management: Bishal Saha
- Presentation Management: Muntasir Mamun Fahim
- Report Compilation: Istiaq Ahmed

Current Progress:

* **DONE** - Identify the header files and core variables (`xw`, `ow`, `dw`) to be used.
* **DONE** - Map out the initial 3x3 board design using a character array (`n[9]`).
* **DONE** - Implement the primary console grid rendering template using formatted `printf` statements.
* **IN PROGRESS** - Structuring the `do-while` game loop and player turn iterations.
* **TO BE DONE** - Implement player move input collection and input validation.
* **TO BE DONE** - Add win/draw condition checking logic.
* **TO BE DONE** - Full program debugging, testing, and final polish.

Conclusion:

We are currently in the core implementation phase of the project. While the visual board rendering and basic loop frameworks have been established, our main focus remains on implementing player input handling, win-checking algorithms, and integrating file/score persistence if required. We continue to collaborate closely and refine our C programming skills through this project.